



MINI PROJECT PRESENTATION

Multi-Agent Research System

A Multi-Agent AI Orchestration System for Automated Desk Research

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The Problem: Desk Research Is Slow and Manual

- Preparing a briefing on an unfamiliar topic follows a fixed sequence: find credible sources, read them, condense them into prose, judge if it is complete.
- That sequence takes 2-4 hours per topic for a human analyst.
- Output quality varies widely with the individual analyst's skill and diligence.
- Capacity scales only by hiring more people — cost rises in direct proportion to volume.
 - Because the sequence is predictable, it is a strong candidate for automation.

Design Goals — Agreed and Numeric

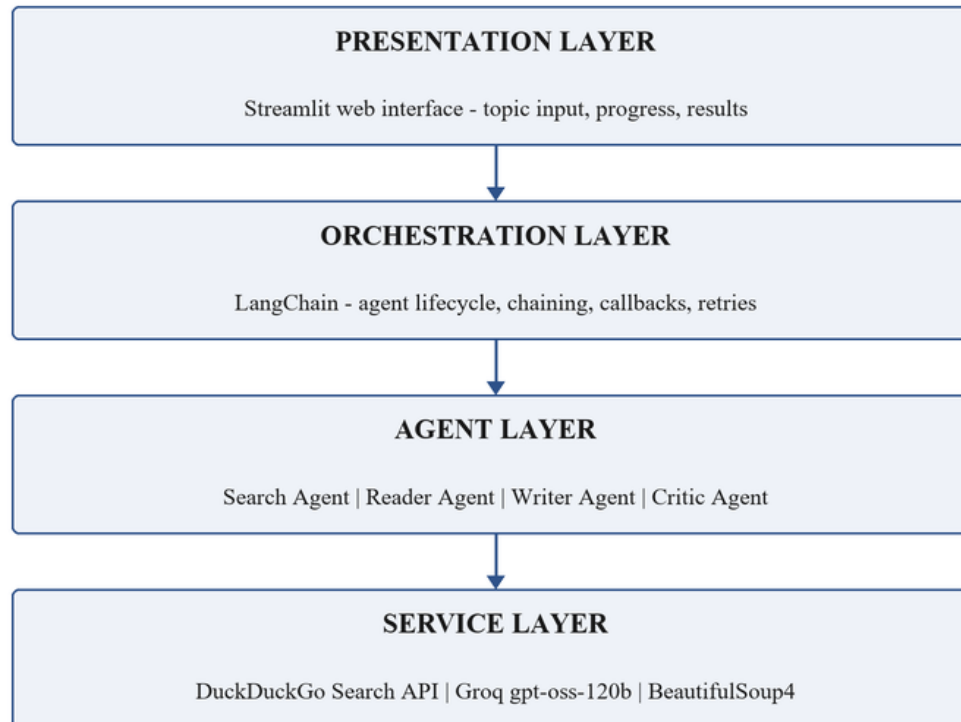
- Perform the whole desk-research sequence with no human intervention.
- Speed — produce a complete report within 60 seconds.
- Cost — substantially below the commercial LLM endpoint then in use.
- Scale — serve at least 10 simultaneous users without degradation.
- Gain practical experience of the engineering around ML work: automated testing, benchmarking and deployment — not model development in isolation.

Success is demonstrated by measurement, not asserted in prose.

Techniques and Tools Studied

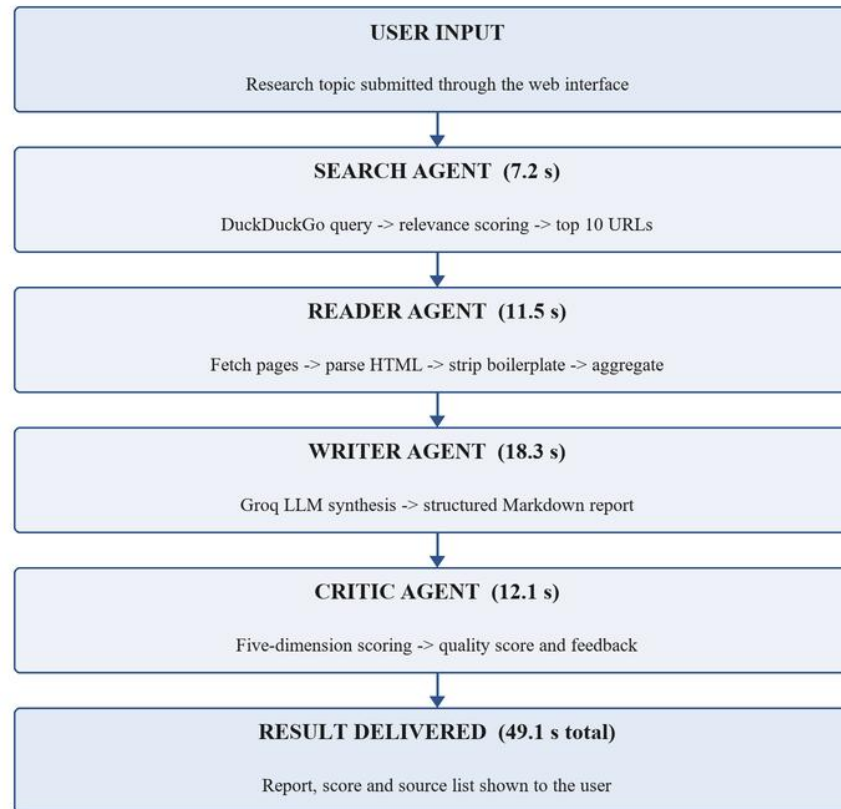
- Large Language Models — stateless; everything the model must consider has to be in the prompt. Instruction phrasing that fixes output shape beats phrasing that asks for quality.
- The Multi-Agent Pattern — split a task into single-purpose stages with declared inputs/outputs, instead of one opaque prompt.
- LangChain — templated prompts, provider abstraction, output parsing, retry behaviour.
- Web retrieval & extraction — Tavily / DuckDuckGo for discovery, BeautifulSoup4 for stripping markup down to body text.
- Streamlit — renders a Python script as a web app; simple UI, no second language or build step.

System Architecture — Four Layers



Presentation → Orchestration → Agents → Services. No layer reaches past its neighbour, so a change in one (e.g. swapping the model provider) does not propagate.

The Four-Agent Sequential Pipeline



Fixed order: reading needs sources, writing needs material, evaluation needs a report. Measured mean end-to-end: 49.1 s against a 60 s ceiling.

What Each Agent Does

Agent	Receives	Returns
Search	Topic string	Up to 10 sources, each with URL, title, snippet, relevance score
Reader	Ranked sources	Aggregated body text + counts of successful / failed retrievals
Writer	Aggregated text + topic	Report in Markdown, six fixed sections
Critic	Report + topic	Score 0-10, five sub-scores, written strengths & weaknesses

Each agent has a declared contract — so each can be timed, tested and fixed on its own, and a defect localises to one stage.

Technology Stack and Rationale

Layer	Technology	Reason for selection
Interface	Streamlit	Simple UI; no second language or build step
Orchestration	LangChain	Prompt templating, output parsing, provider abstraction
Language model	Groq gpt-oss-120b	~70% cheaper per token; very low latency
Discovery	Tavily / DuckDuckGo	No quota at the volumes involved
Extraction	BeautifulSoup4	Tolerant of malformed markup; well documented
Config	python-dotenv	Keeps credentials out of the repository
Testing	pytest	Fixtures and parametrisation suit agent testing

Implementation Highlights

- Prompt design — the Writer prompt names six sections with word budgets; stating structure explicitly was the single biggest gain in output consistency.
- Sampling — temperature 0.7 for synthesis (readable prose), 0.3 for the Critic (consistent scoring).
- Reader is built to fail gracefully — a dead URL removes one source, it does not end the request; extractions under 100 words are rejected.
- Error handling — page failure absorbed by Reader; model-endpoint failure retried with exponential backoff (2 s $\rightarrow$ 32 s); missing credential is terminal.
- Stateless — each request owns its state; no database; any instance can be restarted without data loss.

User Interface

Multi-Agent Research System

Enter a research topic

Research

Stage 3 of 4 - Writer Agent - 31 s elapsed

Generated research report

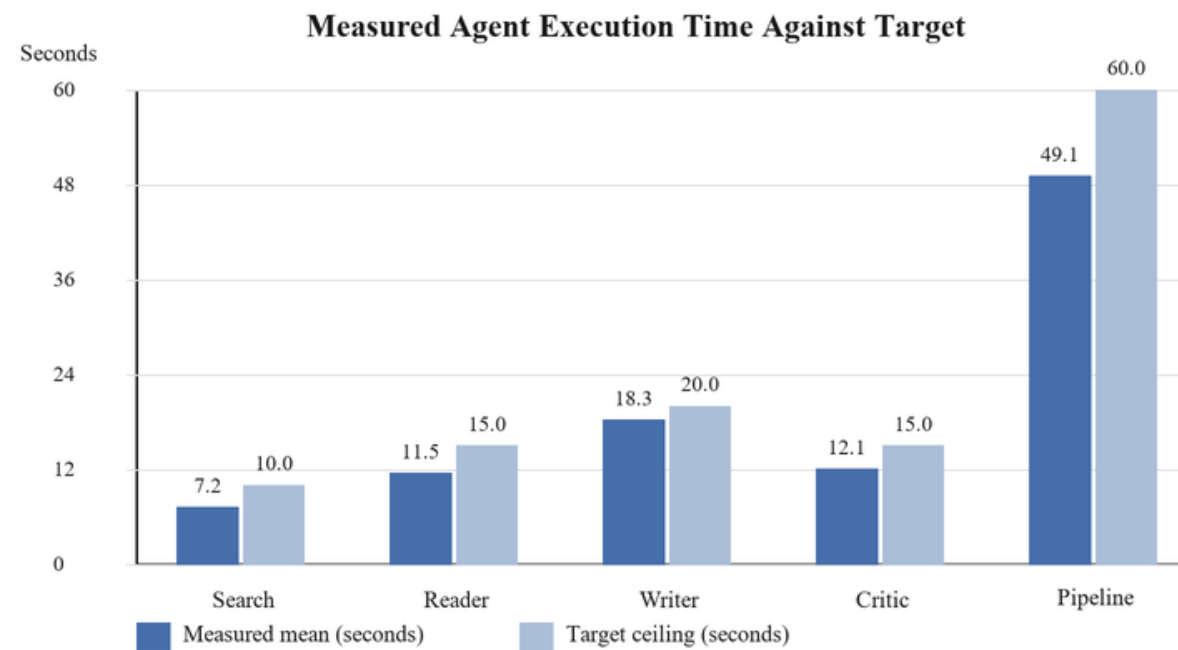
Quality

8.2 / 10

One text field, a button, a live progress indicator, the rendered report with its quality score and every source listed. All generated content is escaped before rendering.

Testing and Performance

- 39 automated tests — unit, integration, end-to-end — all pass.
- Statement coverage: 85% overall (target met).
- Every stage finishes inside its budget.
- Pipeline mean: 49.1 s vs 60 s ceiling; 95th pct 58.1 s.
- 10 concurrent users add only ~1.4 s of latency.



Cost Evaluation

Item	Commercial endpoint	Groq endpoint
Model charges / month	5.25 USD	1.75 USD
Compute instance	30.00 USD	30.00 USD
Storage and transfer	2.00 USD	2.00 USD
Monthly total	37.25 USD	33.75 USD
Cost per report	0.0373 USD	0.0338 USD

Model charges fell ~70% (the target). Effect on the total is smaller because the compute instance dominates at 1,000 reports/month — the saving grows as volume rises, since model cost scales with usage and the instance cost does not.

Known Limitations

- No fact-checking — the system synthesises what its sources assert; wrong sources produce a confident, well-formed, wrong report.
- The Critic scores craftsmanship, not correctness — it has no ground truth.
- Extraction is heuristic — fails on script-rendered pages, paywalls and non-HTML; typically 1-3 of 10 URLs are lost.
- Pipeline is strictly sequential — it cannot revisit an earlier stage in light of a later one.
- English only; nothing stored, so no report history; Critic Agent coverage is 79% vs the 85% target.

Conclusion and Future Scope

- The system performs the whole desk-research sequence unattended and meets every target: 49.1 s mean, ~70% lower model cost, 10 concurrent users, 39/39 tests.
- The decisive decision was four agents over one model — the gain was diagnosability, not just quality. Nearly every measurement depends on it.
- Future scope:
 - Act on the Critic's score — low score triggers a fresh search (line → loop).
 - Better extraction via a headless browser or article-extraction library.
 - Cache results by topic; add multi-language and PDF support.

Thank You

Questions & Discussion

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